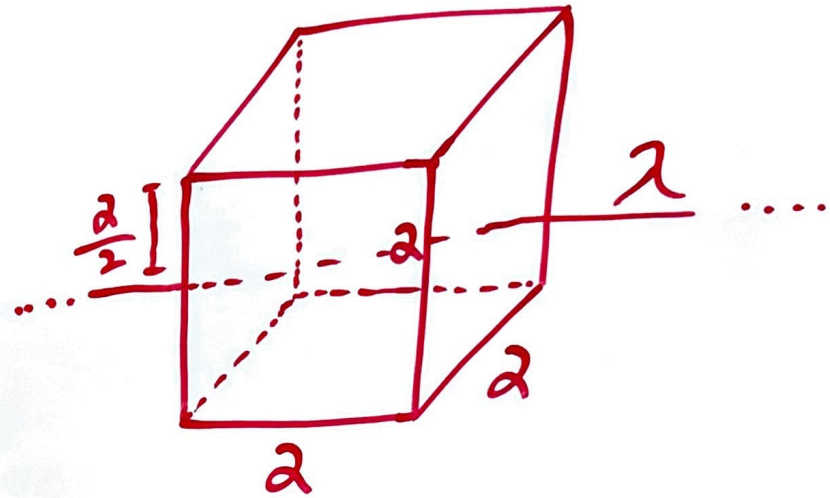
$$\Phi_e = \oint \vec{E} \cdot d\vec{A}$$

$$= \frac{q_{enc}}{\varepsilon_0}$$

$$\text{total flux} = \frac{\lambda a}{\varepsilon_0}$$



surface	flux
left	0
right	0
top	$\frac{1}{4}\frac{\lambda a}{\varepsilon_0}$
bottom	$\frac{1}{4}\frac{\lambda a}{\varepsilon_0}$
front	$\frac{1}{4}\frac{\lambda a}{\varepsilon_0}$
back	$\frac{1}{4}\frac{\lambda a}{\varepsilon_0}$

The area vector of left & right surface are orthogonal to the radially outward electric field, therefore contribute none of the flux.

The other four faces contribute the same amount of flux, therefore the flux value for each face is one forth of the total flux.